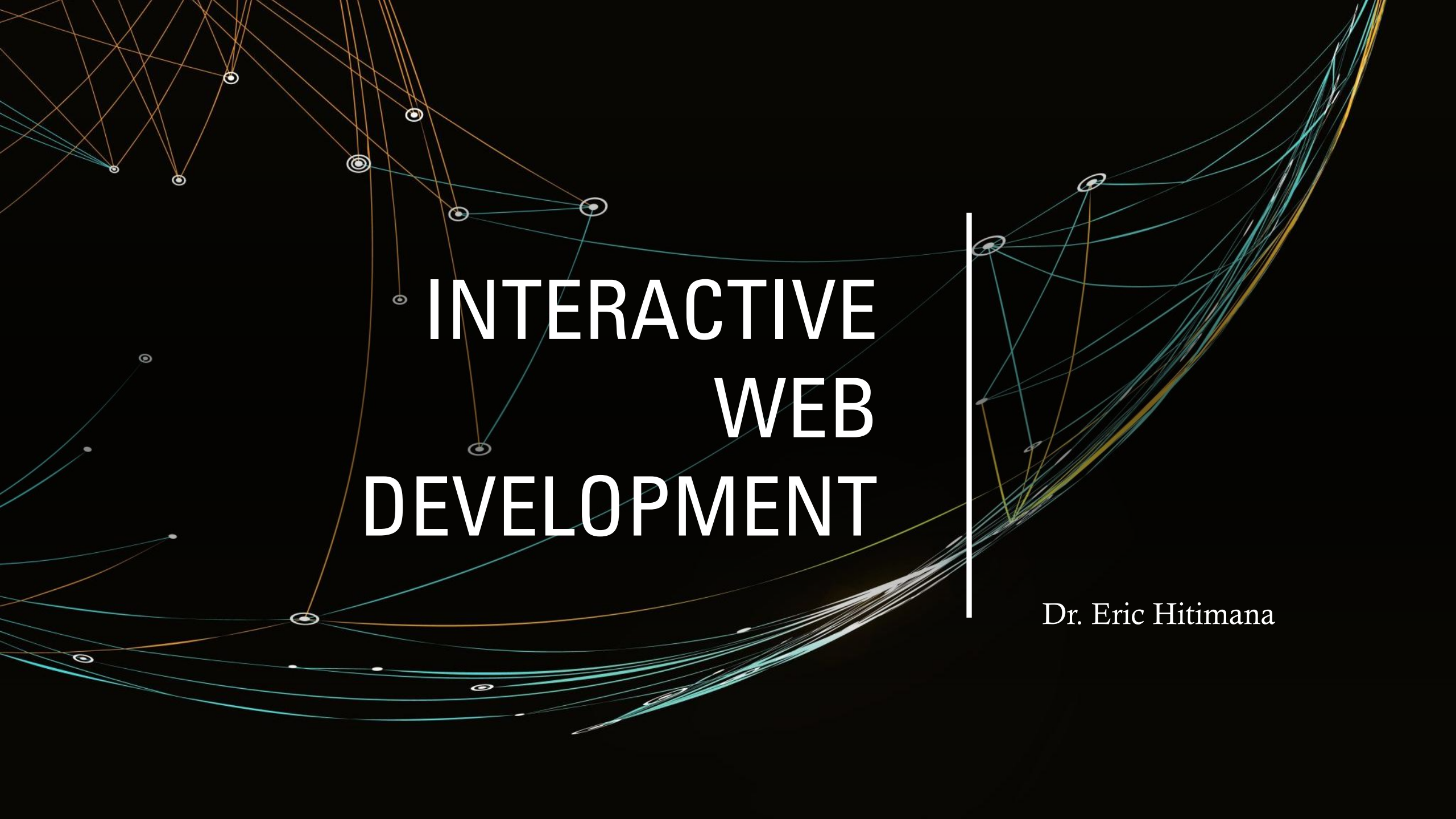


An abstract network diagram with nodes and lines. The nodes are represented by small circles, some of which are white with black outlines, and others are solid black. The lines are thin, curved, and colored in shades of orange, teal, and yellow. They connect the nodes in a complex, web-like pattern. A vertical white line is positioned to the right of the main text.

INTERACTIVE WEB DEVELOPMENT

Dr. Eric Hitimana

Agenda

1. Introduction to Web Development Concepts

- ❖ Intro
 - ❖ UI/UX Design Principles
 - ❖ **Recap on HTML**
 - ❖ **Recap on CSS**
 - ❖ **Recap on JS**
 - ❖ Bootstrap
 - ❖ Project Capstones Recap
-

Introduction

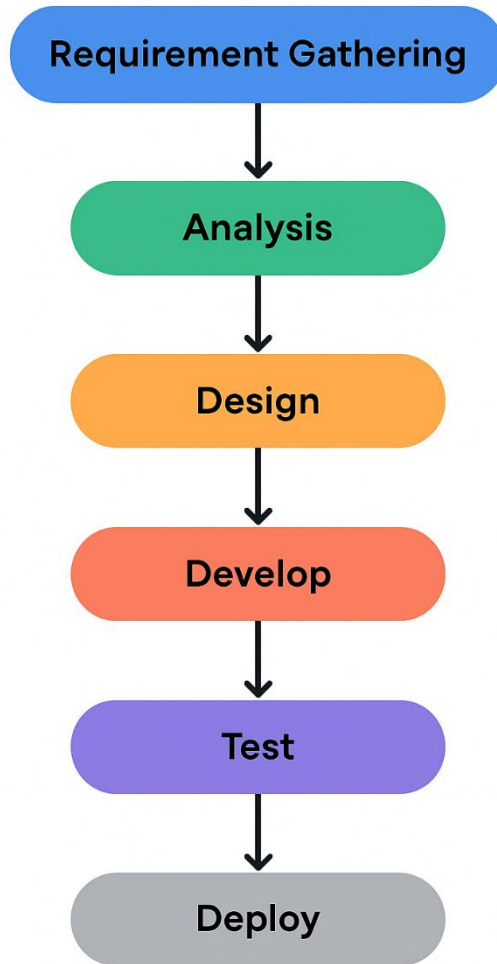
An interactive web application:

- ❖ is a type of website or online platform that goes beyond just displaying static information.
- ❖ It responds to user input and allows users to actively engage with content, perform actions, and receive immediate feedback.

Key points to consider

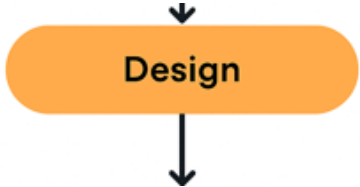
- ✓ **User Engagement:** User clicks buttons, submits forms, drags and drops items ...
 - ✓ ***Real-time feedback:*** Application responds right away to user action without requiring a full page reload. (Online chat, interactive quizzes...)
 - ✓ ***Dynamic Content:*** Content updates based on user interactions or external data sources like APIs. (Weather App).
-

Introduction - Web Development Steps





Design



LECTURE 1:

Recap on UI & UX Design

UI & UX Design – Why It Matters

1. User Interface (UI) – *Look & Feel*

- ✓ Ensures **visual clarity** and **consistent design**
- ✓ Builds **brand trust** through colors, typography, and layout
- ✓ Guides users with **intuitive controls and navigation**

2. User Experience (UX) – *Ease & Flow*

- ✓ Focuses on **user needs, efficiency, and satisfaction**
- ✓ Reduces **errors, frustration, and drop-offs**
- ✓ Enhances **engagement and retention**

3. Business Impact

- ✓ Higher **adoption and loyalty**
- ✓ Reduced **support costs**
- ✓ Improved **ROI**

Good UI catches attention. Great UX keeps users coming back.

UI and UX Design

IDEA THEOREM



Improve Your UI and UX Design for Business

An effective UI/UX design is pivotal in creating terrific interactions for users. A website is only as useful as it is responsive. He or she wants to buy the product or service because of the wonderful user experience they had. Here are a few ways to improve the UI and UX design with ease.





#1 Stick to a Single Typeface

When there is more than one typeface, the user spends time adjusting to the difference. This will make the person lose interest in your product.

#2 Improve User's Onboarding

This is the most important thing. If you use the Skip this Step button to help users avoid cumbersome steps, more customers will look forward to using your UI/UX interface.



#3 Copy Top Designs

Scan the magazines and the images of online ads for hints on how to improve your design. Sometimes, it is obvious.

#4 Get a Mentor

Use the help from a senior UI designer in your office to get tips on how to improve your work. It only takes a few minutes and you will improve the quality of your work.

#5 Ensure Uniformity of Light Source for Shadow

Readers feel comfortable when the shadow is uniform. If there is a disparity, they will spend time wondering what is wrong. This will disrupt the attention they pay to the content.



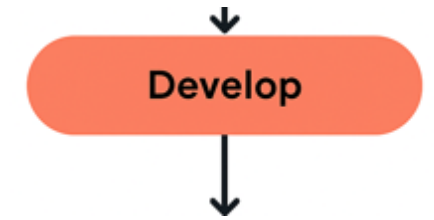
TOOLS for Wireframes/Mockups

1. Pen and Paper
2. Figma - <https://www.figma.com/>
3. PS
4. Draw.io
5. Balsamiq
6. Etc

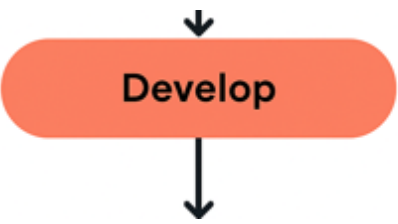




LECTURE 2:



Web Language	Primary Use	Common Frameworks / Libraries (by type)	
 JavaScript	Front-end & Back-end programming	Front-end (Client-side)  React  Angular  Vue.js  Next.js	Back-end (Server-side)  Express.js  NestJS  Fastify
		Front-end (Client-side)  React  Angular  Vue.js  Next.js	Back-end (Server-side)  NestJS  Express.js
 TypeScript	Typed JavaScript development	Front-end (Client-side)  React  Angular  Vue.js  Next.js	Back-end (Server-side)  NestJS  Express.js
 Python	Web applications, APIs, data-driven web apps	Back-end (Server-side)  Django  Flask  FastAPI  Pyramid	
 PHP	Dynamic websites, web systems	Back-end (Server-side)  Laravel  Symfony  CodeIgniter  CakePHP  Zend Framework	
 Java	Enterprise applications, APIs	Back-end (Server-side)  Spring Boot  Jakarta EE  JSF	
 C#	Enterprise applications, APIs	Back-end (Server-side)  ASP.NET Core  Blazor	



~~HTML, CSS, JS,~~ BOOTSTRAP

BOOTSTRAP

Bootstrap is a popular, open-source front-end framework used to design responsive and mobile-first websites.

-
- Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development.
 - It contains HTML, CSS, and JavaScript-based design templates for [typography](#), [forms](#), [buttons](#), [navigation](#), and other [interface components](#).
-

| How to Use |

- A basic understanding of HTML is required to start learning Bootstrap. Some familiarity with how CSS works (CSS Selectors and Visual Rules) would be helpful,
 1. Download Bootstrap from getbootstrap.com
 2. Include Bootstrap from a CDN (Content Delivery Network)

| Bootstrap Grid |

The Bootstrap grid system has five classes:

- ✓ **xs** (<576px wide)
- ✓ **sm** (>576px wide)
- ✓ **md** (>768px wide)
- ✓ **lg** (>992px wide) xl (>1200px wide)
- ✓ **XXL** (For >1400px wide)

[illegible]

| Viewport |

- The viewport is the user's visible area of a web page. It varies with the device and is smaller on a mobile phone than on a computer screen.

`<meta name="viewport" content="width=device-width, initial-scale=1.0">`



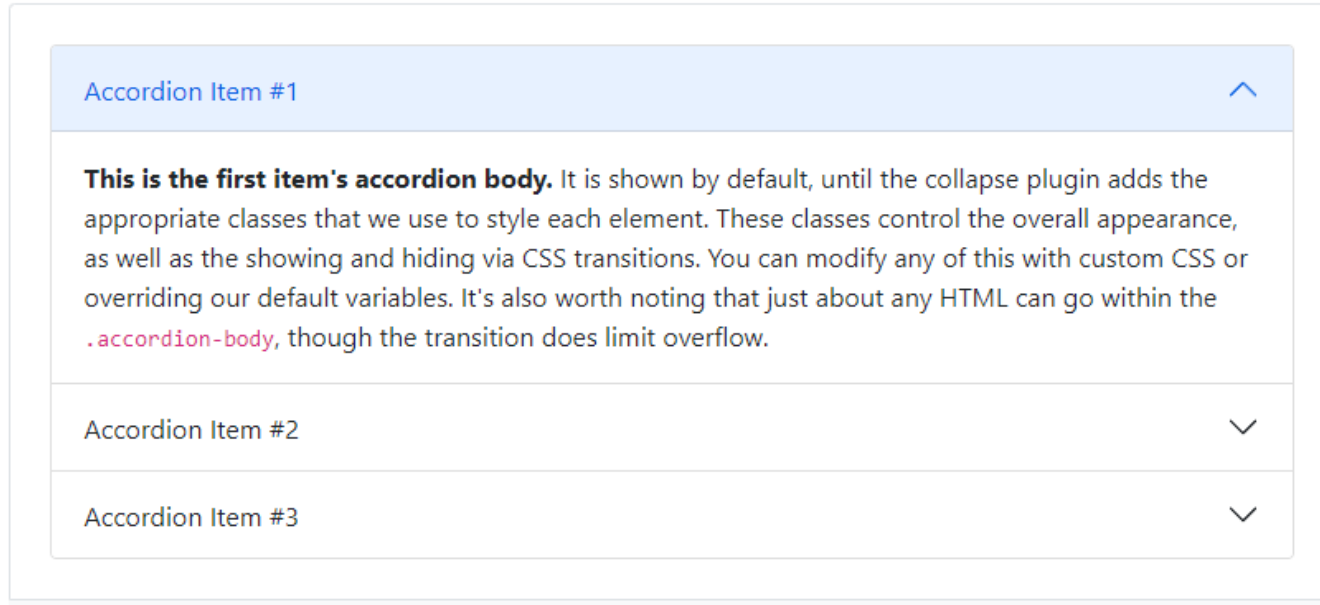
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut aliquip ex ea commodo consequat. Duis autem vel eum iriure dolor in hendrerit in vulputate velit esse molestie consequat, vel illum dolore eu feugiat nulla facilisis at vero eros et accumsan et iusto odio dignissim qui blandit praesent luptatum zzril delenit augue duiis dolore te feugait nulla facilisi. Nam liber tempor cum soluta nobis eleifend option congue nihil imperdiet domo el...

| Components |

- Many more reusable components are built to provide iconography, dropdowns, input groups, navigation, alerts, and much more.
 - Accordion
 - Alerts
 - Badge
 - Breadcrumb
 - Buttons
 - Button group
 - Card
 - Carousel
 - Close button
 - Collapse
 - Dropdowns
 - List group
 - Modal
 - Navs & tabs
 - Navbar
 - Offcanvas
 - Pagination
 - Popovers
 - Progress
 - Scrollspy
 - Spinners
 - Toasts
 - Tooltips
-

| Components |

1. Accordion



```
<div class="accordion" id="accordionExample">
  <div class="accordion-item">
    <h2 class="accordion-header" id="headingOne">
      <button class="accordion-button" type="button" data-bs-toggle="collapse" data-bs-target="#collapseOne" data-bs-expanded="true">
        Accordion Item #1
      </button>
    </h2>
    <div id="collapseOne" class="accordion-collapse collapse show" aria-labelledby="headingOne" data-bs-parent="#accordionExample">
      <div class="accordion-body">
        <strong>This is the first item's accordion body.</strong> It is shown by default, until the collapse plugin adds the appropriate classes that we use to style each element. These classes control the overall appearance, as well as the showing and hiding via CSS transitions. You can modify any of this with custom CSS or overriding our default variables. It's also worth noting that just about any HTML can go within the .accordion-body, though the transition does limit overflow.
      </div>
    </div>
  </div>
  <div class="accordion-item">
    <h2 class="accordion-header" id="headingTwo">
      <button class="accordion-button collapsed" type="button" data-bs-toggle="collapse" data-bs-target="#collapseTwo">
        Accordion Item #2
      </button>
    </h2>
    <div id="collapseTwo" class="accordion-collapse collapse" aria-labelledby="headingTwo" data-bs-parent="#accordionExample">
      <div class="accordion-body">
        <strong>This is the second item's accordion body.</strong> It is hidden by default, until the collapse plugin adds the appropriate classes that we use to style each element. These classes control the overall appearance, as well as the showing and hiding via CSS transitions. You can modify any of this with custom CSS or overriding our default variables. It's also worth noting that just about any HTML can go within the .accordion-body, though the transition does limit overflow.
      </div>
    </div>
  </div>
  <div class="accordion-item">
    <h2 class="accordion-header" id="headingThree">
      <button class="accordion-button collapsed" type="button" data-bs-toggle="collapse" data-bs-target="#collapseThree">
        Accordion Item #3
      </button>
    </h2>
    <div id="collapseThree" class="accordion-collapse collapse" aria-labelledby="headingThree" data-bs-parent="#accordionExample">
      <div class="accordion-body">
        <strong>This is the third item's accordion body.</strong> It is hidden by default, until the collapse plugin adds the appropriate classes that we use to style each element. These classes control the overall appearance, as well as the showing and hiding via CSS transitions. You can modify any of this with custom CSS or overriding our default variables. It's also worth noting that just about any HTML can go within the .accordion-body, though the transition does limit overflow.
      </div>
    </div>
  </div>
</div>
```

| Components |

2. Alerts

A simple primary alert—check it out!

A simple secondary alert—check it out!

A simple success alert—check it out!

A simple danger alert—check it out!

A simple warning alert—check it out!

A simple info alert—check it out!

A simple light alert—check it out!

A simple dark alert—check it out!

```
<div class="alert alert-primary" role="alert">
  A simple primary alert—check it out!
</div>
<div class="alert alert-secondary" role="alert">
  A simple secondary alert—check it out!
</div>
<div class="alert alert-success" role="alert">
  A simple success alert—check it out!
</div>
<div class="alert alert-danger" role="alert">
  A simple danger alert—check it out!
</div>
<div class="alert alert-warning" role="alert">
  A simple warning alert—check it out!
</div>
<div class="alert alert-info" role="alert">
  A simple info alert—check it out!
</div>
<div class="alert alert-light" role="alert">
  A simple light alert—check it out!
</div>
<div class="alert alert-dark" role="alert">
  A simple dark alert—check it out!
</div>
```

| Advantages of Bootstrap |

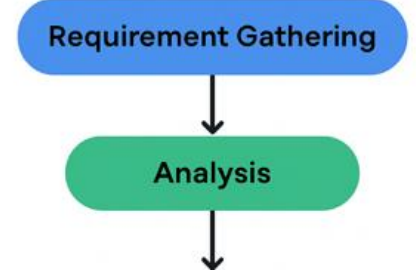
1. Fewer cross-browser bugs
2. A consistent framework that supports the majority of all browsers and CSS compatibility fixes,
3. Lightweight and customizable,
4. Responsive structures and styles,
5. Several JavaScript plugins using the jQuery,
6. Good documentation and community support,
7. Loads of free and professional templates, WordPress themes, and plugins
8. Great grid system

| Disadvantages of Bootstrap |

1.
2.
3.
4.
5.

PROJECT BRAINSTORMING AND TAKE- HOME

1. Teamwork of 2 members
2. Decide the domain (...)
3. Think of a problem to be solved by IT solution
4. Ensure it has the components below (WEB, MOBILE, AI, and EMBEDDED SYSTEM).
5. **WORK #1:** Submit the project title and description of the problem and the right client to benefit from the solution by tomorrow, 5 pm. Make sure the Figma Design is ready by Next Monday at 1 PM.



YEAR 1 WEB PROJECT TITLES

Project repository review and consolidated project list

No.	Student Name	Reg. Number	Project Title
1	Abayisenga Ndekezi Bertin	225084736	ProjectHub – Student-Investor Project Marketplace
2	KWIZERA IRAGUHA Yann	225061020	Student Project Marketplace and Collaboration Platform
3	NIYOMWUNGERI Sandrine	225054717	Fresh Saver – Food Freshness and Management Platform
4	MUKAMISHA ESTHER	225071210	Smart Rent – Online Property Rental Platform
5	NKUSI NGENZI Chaste	225056357	Healthcare Appointment Management System
6	Jules MUHIRWA	225067392	SPIMAS – Smart Public Infrastructure Monitoring & Accountability System
7	NIYONGOMA Musare Pacifique	225062054	Digital Mining Information and Management Platform
8	ISHIMWE DAVID BOB	225058590	Travo – Rwanda Travel and Tourism Platform
9	Daniella Ishimwe	225078324	Waste Management Information System
10	Gihozo Ange Chanella	225059939	University Lost & Found Portal
11	MUTUYIMANA PACIFIQUE	225031462	Ubumenyi Hub – Digital Learning Platform
12	IRAKOZE Arsene	225033151	Flash Cards AI Maker – AI-Powered Learning Platform
13	Amiel NIYOGISUBIZO	225061402	AgriConnect Rwanda – Digital Agricultural Marketplace

14	Mugisha Elise	225059498	Smart Finance – Personal Financial Management System
15	Izihirwe Harerimana Jean Arcade	225033615	Hospital Appointment Management System
16	Iryamukuru Ham Odhon	225056635	UMURANGA – Skilled Workers and Services Platform
17	BAMBE Kwizera Jean Paul	225060001	AssistAbility – Disability Support and Inclusion Platform
18	NIYOMWUNGERI FEBRONIE LOUANGE	225063489	Lost and Found Web Application
19	SHIMO SEZERANO Second Adelin	225055530	SmartUmuganda – Digital Civic Reporting Platform
20	Jean Claude MUGISHA	225057478	UBUZIMA – Youth Health Information and Quiz Platform
21	IHIMBAZWE Placide	225032165	Simpli Motors – Vehicle Sales and Management Platform
22	SANGO NCUNGU Blaise Joy	225058527	Elite Shop – Online Technology Products Store
23	NISENGWE Joyeuse	225059281	Choir Management and Membership Platform
24	MUKUNDABANTU MOHAMMED	225029152	Web Design Learning and Practice Platform
25	ABIJURU KUBWAYO Yvan	225028760	Farm-to-Table Agricultural Marketplace
26	Barasingiza Nzaramba Ismael	225043521	Web Design Project

27	Fidele Ihorindengera	225039978	Online Hospital Appointment Booking System
28	MBUTO MUGISHA Bruce	225043452	Tugende – Bus Ticket Booking System
29	IZERE Abijuru Cprien	225030699	Unique Hotel – Hotel Booking and Information Platform
30	Manzi Olivier	225064165	Digital Umuganda Attendance and Accountability System
31	Umukunzi David	225061423	FoodBox – Fresh Ingredients and Recipe Platform
32	NTIRANDEKURA Daniel	225064517	School Resource Tracking System
33	Umurerwa Anne Christella	225033533	SafeReport – Community Issue Reporting Web Application
34	BARAME Leila Martine	225038140	Smart Workforce Management System
35	Jules MUHIRWA	225067392	SPIMAS – Smart Public Infrastructure Monitoring & Accountability System (HTML Version)
36	ISHIMWE Manzi Samuel	225032016	Agritera – Agricultural Information and Support Platform
37	HIRWA Benjamin	225041181	IKODE – Online Rental Agency Platform
38	GIHOZO TETA Divine	225051948	Rwanda Aviation Hub – Aviation Information Platform
39	IKUZO BENOIT ARNAUD	225050216	APFTS – AI-Powered Public Funds Transparency System

40	Ndikubwimana Didier Gilbert	225058410	Smart Land Management System
41	MUNEZERO Arael	225058843	Web-Based HTML Design Project
42	Nyamuberwa Wilson	225059015	Komeza Umutekano – Safety and Security Management System
43	TUYISHIMIRE HIRWA Robert	225060097	Shimwa Design – Smart Tailor Management and Fashion Platform
44	NIYOKWIRINGIRWA Ismael	225032223	Agri-Link Rwanda – Farmer-to-Buyer Agricultural Platform
45	JAMBO SONGA Aime Serge	225032946	Agri-Nexus – Agricultural Information and Market Platform
46	Bigira Josue	225040993	Ibiciro Rwanda – Agricultural Market Price Information System
47	Mumporeze Teta Christa	225072339	AgriLink – Agricultural Marketplace and Farmer Support Platform
48	IRAKOZE INGABIRE Aphia	225067917	Twizere Cybernet – Cybersecurity Web Platform
49	Flora Ishimwe	225031794	Cybersecurity Learning Platform
50	IGENO Olivier Fidele	225036732	Rwanda Nziza – Citizen Public Problem Reporting Platform
51	Nambajimana Gratien	225031064	Kid Learning Hub – Digital Learning Platform for Children
52	Mugabwambere Dieudonne	225052665	Web-Based Image Recognition System

53	NIWEMUTONI Emelyne	225074534	UniBridge – University Student Services Platform
54	UMUKUNZI Sano Arsene	225067539	MediLink – Healthcare Operations and Digital Patient Records Platform
55	Rwema Eulade	225032363	e-LinkAcre – Farmer, Buyer and Government Agricultural Platform
56	Fulgence NIYOMUKIZA	225056492	eHealth Hub – Healthcare Management and Appointment System
57	Kwizera Salvator	225070444	CampusSync – Online Learning Platform
58	RUGIRA Eric	225037467	QuickQue – Queue Management System
59	Twambazimana Elie	225036380	Assignment Management System
60	Muhire Herve	225061404	Hospital Management System
61	MUHORAKEYE Alice	225062496	Online Accommodation Booking Platform
62	KWIZERA Merveux	225031550	Arsenal Fans Website
63	UMWANAMWIZA SERGE	225045359	BizFlow Rwanda – SME Business Management System
64	ISHIMWE Niyoyita Kevin	225074650	Smart Agriculture Web Application
65	IHIMBAZWE MUGENZI DENYS	225047513	HealthConnect – Healthcare Management Platform

❑ TOPIC: Smart Seeds Nursery and Distribution Management, Case of Karongi District

❑ Group Members:

1. 2012444

2. 2021393

PROBLEM

- ☐ Lack of ready seeds localization
- ☐ Seeds category distribution
- ☐ Lack of seeds company categorization per seed type
- ☐ Monitoring of seed consumers for future decision making
- ☐ Waste of water for irrigation practice
- ☐ Lack of Seeds with fertilization type mapping

FEATURES

Responsive WEB APP

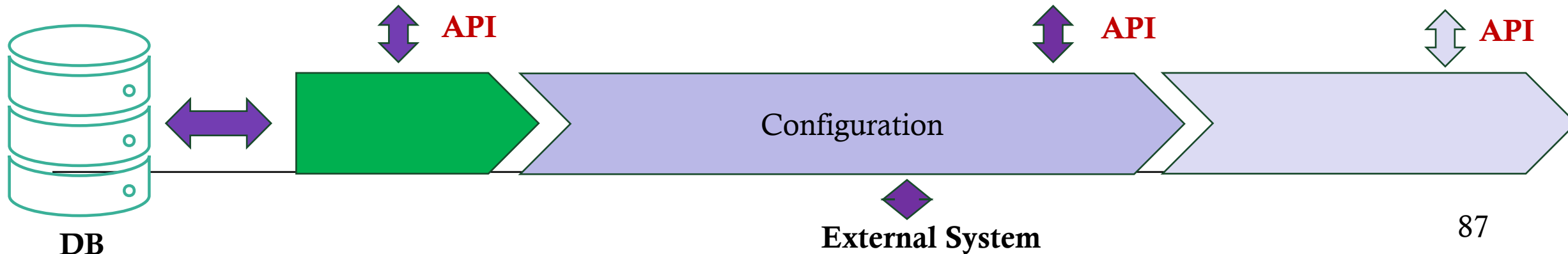
- User authentication & Authorization
- Localization
- User role Management
- Cooperative Management
- Plot/Farm management
- Seed category Management
- Seed Management
- Localization management
- Fertilizer Category Management
- Fertilizer Management
- Device (IoT) Management
- Device Allocation vs status Management
- Data Reporting Management
- Data Visualization Management

Mobile APP

- Authentication
- Monitor the farm
- Access ready seeds with details
- Book ready seeds
- Access irrigation schedule using AI
- Offline mode capability

IoT Device

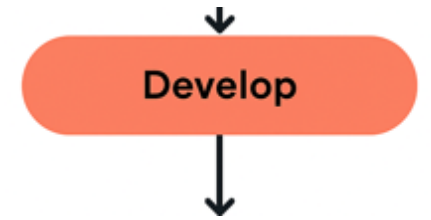
- ✓ Power management
- ✓ Data acquisition
- ✓ Data submission



USERS

- ☐ Individual Farmer
- ☐ Cooperative Manager
- ☐ Local government leaders
- ☐ Partners (NAEB)
- ☐ Admin

PRACTICAL ON BOOTSTRAP



1. Download your convenient free template and customize it like your FIGMA context.
2. **WORK #2:** Progressive evaluation and marking will be conducted.

Q/A

Feedback
